

# Assessment of the Level of Software Development Using CMMI-Dev 1.3 on the Unjani Official Web

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## Abstract

UNJANI website ranked 186 in Webometric ranking. Webometric is a ranking system of universities around the world conducted by Cybermetrics Labs which is useful to determine the ranking in terms of the level of performance indicators of universities, which include Visibility, Openness, Excellence. The purpose of this study was to assess the level of maturity of the software development process on the UNJANI website using CMMI Dev 1.3. This study uses quantitative methods, with a survey approach to the person in charge and manager of the UNJANI website. Quantitative research is a way to obtain research data measured using tools such as surveys, questionnaires, or experiments, and analysed with statistical methods to find certain patterns, relationships, or trends. The results of this study found that out of 7 CMMI Dev.1.3 process areas, some areas are in accordance with the standard process of making the UNJANI website and some process areas are not in accordance with the CMMI Dev 1.3 framework. The maturity level of the UNJANI website is in the category of partially implemented in most process areas, with the highest GAP (50%) in Configuration Management (CM) and the lowest GAP (30%) in Project Monitoring and Control (PMC). The conclusion of this research reveals that UNJANI website still needs improvement in configuration management, change recording, and feature quality evaluation before release.

**Keywords:** CMMI-DEV 1.3, GAP Analysis, SCAMPI Class C, UNJANI Website, Webometrics

## Introduction

Webometrics is an assessment system that uses university websites to assess the performance of all universities around the world [1], [2], [3]. This system uses four criteria or indicators, namely presence, visibility impact, openness, and excellence. Each of these indicators has a different measure to be assessed. The openness indicator, which includes the number of Adobe Acrobat (.pdf), Adobe PostScript (.psd), and other document files, has a weight of 10%. The presence or presence indicator, which counts each college web page indexed by search engines, has a weight of 50%. The visibility indicator, or visibility impact, has a weight of 50%. On the Webometric Ranking. Universitas Jenderal Achmad Yani is ranked 186th in Indonesia out of 3,523 Indonesian universities, when compared with several private universities with the same number of students, UNJANI is ranked low [4]. In this case, there are several things that will be traced using the CMMI Dev 1.3 method assessment.

Abdul Barir Hakim conducted a research [5]; due to the relatively small size of the company, this research discusses the application of CMMI to software development with continuous representation. This research uses SCAMPI C for the measurement process, which does not require as many human resources as SCAMPI A and SCAMPI B. The result is that the company must make policies,

which will be used for the measurement process. The result is that the company should create policies, procedures, and documentation templates in accordance with CMMI standards.

Furthermore, Kurniawati's research used a staged representation. Thus, the process areas evaluated are process areas that exist at maturity level 2 and are connected to the CMMI-Dev 1.3 structure [6]. The evaluated process areas are also process areas that exist at maturity level 2. This research also provides recommendations for improving project management.

UNJANI's websites and webometrics determine the ranking of institutions in a university, which measures the visibility and digital impact of a university through online-based university information. The existing information technology assets at Unjani have a high risk, which has an impact on the sustainability of the use of information technology. This is because information technology governance has not been carried out constructively [7]. Key indicators such as presence (number of indexed web pages), visibility (number of external links to university domains), openness (number of publicly accessible document files), and excellence (number of frequently cited scientific publications) all depend on the quality and management of website software. Websites supported by software with a high level of maturity enable better management of content, configuration and documentation, thereby improving performance in these indicators. In the context of UNJANI, to understand the extent to which the software management process contributes to the ranking in Webometrics.

In previous research references related to measuring the extent to which application software development supports Manans's Farm business process activities and the maturity level of the Manans MINT application from the software assessment process using CMMI-Dev 1.3 have been carried out with the results [8]. In other research, discussing the maturity level and capability of the software development process in VTD (virtual team development) [9], as well as providing strategies for companies that use Outsourcing Virtual Team Development services. in developing quality software.

## **Material and Methods**

### **A. Research Design**

This study adopts the staged representation model of CMMI-DEV 1.3 as the primary framework to evaluate the maturity of existing processes, focusing specifically on Maturity Level 2 (Managed) and Level 3 (Defined) [10], [11], [12]. These levels were selected because they represent the foundational stages for establishing controlled and documented processes within an organisation. To support this evaluation, the SCAMPI Class C method is employed due to its flexible, low-cost, and diagnostic nature, making it suitable for academic research. SCAMPI Class C allows researchers to assess process performance without the formal requirements of higher-class appraisals while still generating meaningful insights regarding process capability and compliance with CMMI standards. The subjects of this research include individuals responsible for managing and developing UNJANI websites, consisting of the website coordinator and the software development team.

### **B. CMMI Dev 1.3**

CMMI Dev 1.3 defines organizational maturity through five progressive levels that represent the consistency, predictability, and effectiveness of a company's processes [10], [12]. The highest maturity

level, Level 5, reflects an organisation that has fully optimised its processes and continuously improves them through quantitative insights and innovation-driven practices. Within this model, a key process area (KPA) represents a group of related activities performed collectively to achieve specific organisational objectives [13]. Each KPA is associated with a set of goals that must be accomplished to demonstrate process capability at a given maturity level. Successful achievement of these goals serves as evidence of the organisation's current maturity status [14].

To guide the implementation of each KPA, CMMI introduces common features, which function as structured instructions for carrying out critical process elements. These features consist of five components: commitment to perform, ability to perform, activities performed, measurement and analysis, and verification of implementation. Together, they ensure that process areas are executed systematically and sustainably. Additionally, key practices describe the essential actions or techniques required to implement each process area effectively [15].

. These practices provide detailed operational guidance, enabling organisations to institutionalise high-quality processes and achieve continuous improvement across development activities.

### **C. SCAMPI Class C**

A SCAMPI Class C appraisal is one of three appraisal types within the Standard CMMI Appraisal Method for Process Improvement (SCAMPI). Compared to the more rigorous Class A and Class B appraisals, Class C is the least formal and is primarily used as an initial diagnostic tool to understand an organisation's current process capability [16]. This type of appraisal focuses on providing rapid, flexible feedback rather than producing formal ratings or certifications [17]. Because of its lightweight structure, a Class C appraisal is commonly conducted early in a process improvement initiative to identify strengths, weaknesses, and potential gaps across key process areas. Organisations use the findings to prioritise improvement actions, refine existing practices, and prepare for more formal appraisals in the future. Overall, SCAMPI Class C serves as an efficient and cost-effective way to gain insights into process performance and guide continuous improvement efforts [18].

### **D. System Usability Scale (SUS)**

The System Usability Scale (SUS) is a widely used, standardised instrument for evaluating the usability of systems such as websites, applications, and software tools. SUS provides a quick and reliable way to capture users' perceptions through a ten-item questionnaire with five response options ranging from "strongly agree" to "strongly disagree" [19], [20]. This method is valued for its simplicity, flexibility, and effectiveness across various domains, making it suitable for both formative and summative usability assessments. SUS generates a single score from 0 to 100, which reflects the overall usability of the system. The score can be used to classify usability levels, compare different system versions, or benchmark performance against industry standards. Because of its efficiency and robustness, SUS is commonly applied in usability testing, product evaluations, and user experience research to support decision-making and guide further system improvements [21].

### **E. Likert scale**

The Likert scale is a commonly used instrument for gathering quantitative and qualitative data related to individuals' opinions, perceptions, and attitudes. It presents respondents with a series of

statements and asks them to indicate their level of agreement or disagreement [22]. Through this structured format, the Likert scale helps researchers measure subjective viewpoints in a consistent and interpretable manner, making it valuable for surveys, evaluations, and social science research.

Table 1 provides the Likert scale used in the questionnaire, outlining five response options ranging from “Never” to “Always.” This scale helps quantify respondents’ perceptions and frequency of practice implementation, enabling consistent and structured data interpretation across all evaluated process areas.

**Table 1.** Likert Scale

| Skala | Description |
|-------|-------------|
| 1     | Never       |
| 2     | Rarely      |
| 3     | Sometimes   |
| 4     | Often       |
| 5     | Always      |

#### F. Gap Analysis Concept

Gap analysis is a method used to compare the actual performance of a variable with consumer expectations to identify differences or gaps. This approach helps determine the level of conformity between what customers expect and what a company delivers, thereby indicating overall customer satisfaction [23]. By highlighting discrepancies, gap analysis provides insights into areas where performance falls short and requires improvement. It also reflects how well a service provider understands customer needs and how effectively those needs are being met [24]. As a result, gap analysis serves as a valuable tool for enhancing service quality and guiding strategic decision-making.

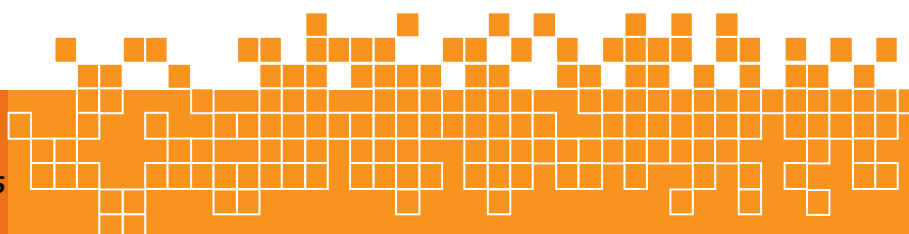
#### G. Experimental Setup

Data were gathered using a Likert scale-based questionnaire (1–5) distributed to respondents, including the person in charge and members of the development team. The instrument was designed to assess perceptions and performance across seven key process areas relevant to project and process management. Each respondent rated statements reflecting the implementation level of these areas, enabling quantitative analysis of organisational strengths and gaps. The collected data served as the basis for evaluating process maturity and identifying priority improvements within the development environment.

Table 2 lists the key process areas evaluated in the experimental setup, along with their corresponding abbreviations. These areas represent essential components of the CMMI-DEV framework, covering planning, requirements, configuration, quality assurance, monitoring, measurement, and supplier management, forming the foundation for assessing process maturity.

**Table 2.** Key Process Areas Evaluated in the Experimental Setup

| No. | Key Process Area                      | Abbreviation |
|-----|---------------------------------------|--------------|
| 1   | Project Planning                      | PP           |
| 2   | Requirements Management               | REQM         |
| 3   | Configuration Management              | CM           |
| 4   | Process and Product Quality Assurance | PPQA         |
| 5   | Project Monitoring and Control        | PMC          |
| 6   | Measurement and Analysis              | MA           |
| 7   | Supplier Agreement Management         | SAM          |



**H. Data Analysis**

The SCAMPI Class C method is employed as the primary evaluation framework to identify the gap between the actual condition of website management and the ideal standards defined by CMMI-DEV 1.3. This approach allows for a lightweight yet systematic assessment of process performance across the seven key process areas. Data collected through the Likert scale-based questionnaire are analysed by first calculating the average score for each process area, representing the respondents' perception of implementation effectiveness. These average scores are then compared against predefined criteria to determine the corresponding maturity level category: Fully Implemented, Partially Implemented, or Needs Improvement. Through this process, the analysis highlights which areas meet expected standards and which require targeted improvements. The resulting gap information provides valuable insights for refining development processes, prioritising corrective actions, and planning future process enhancement initiatives aligned with CMMI best practices.

Table 3 outlines the parameters assessed for the person in charge across three key process areas. Each area includes specific measured elements: Project Planning evaluates documentation, resource planning, and risk management; PPQA assesses quality standards, feature testing, and evaluations; while PMC focuses on monitoring, progress documentation, and corrective actions.

**Table 3.** Person in Charge Parameter

| No | Process Area                                 | Measured Parameters            |
|----|----------------------------------------------|--------------------------------|
| 1  | Project Planning (PP)                        | Documentation Project Planning |
|    |                                              | Resource Planning              |
|    |                                              | Risk Management                |
| 2  | Process and Product Quality Assurance (PPQA) | Project quality standards      |
|    |                                              | feature testing                |
|    |                                              | regular quality evaluation     |
| 3  | Project Monitoring and Control (PMC)         | Project monitoring             |
|    |                                              | progress documentation         |
|    |                                              | corrective action              |

Table 4 presents the parameters evaluated for the developer team across key process areas. REQM focuses on requirement logging, change management, and reviews; CM assesses configuration changes and validation; PPQA examines development quality and testing; MA measures data management practices; and SAM evaluates the establishment of supplier agreements.

**Table 4.** Developer Team Parameters

| No | Process Area                                 | Measured Parameters                       |
|----|----------------------------------------------|-------------------------------------------|
| 1  | Requirements Management (REQM)               | User Requirements Logging                 |
|    |                                              | Requirements Change Management            |
|    |                                              | Requirements Review Before Implementation |
| 2  | Configuration Management (CM)                | Configuration Change Logging              |
|    |                                              | Change Validation                         |
|    |                                              | Authorised Logging                        |
| 3  | Process and Product Quality Assurance (PPQA) | Quality Standards In Development          |
|    |                                              | Module Testing                            |
|    |                                              | Regular Quality Evaluation.               |
| 4  | Measurement and Analysis (MA)                | Building And Managing Measurements        |
| 5  | Supplier Agreement Management (SAM)          | Establishing Supplier Agreements          |

## Results and Discussions

### A. Questionnaire Results and SCAMPI Analysis

Table 5 presents the SCAMPI Analysis Score categories used to evaluate the implementation level of each process area. Scores of 4.0 or higher indicate a Fully Implemented process, demonstrating strong alignment with CMMI requirements. Values between 3.0 and 3.9 reflect Partially Implemented processes, where key practices exist but require refinement. Scores from 2.0 to 2.9 are classified as Less Implemented, showing limited compliance. Values 1.9 or below indicate Not Implemented, meaning essential practices are largely absent.

**Table 5.** SCAMPI Analysis Score

| Value      | Description           |
|------------|-----------------------|
| $\geq 4.0$ | Fully implemented     |
| 3.0 – 3.9  | Partially implemented |
| 2.0 – 2.9  | Less implemented      |
| $\leq 1.9$ | Not implemented       |

The average score for each process area was calculated using the formula in Equation 1.

$$\text{Average} - \text{Average PA} = \frac{\sum(\text{Respondent Score for PA})}{\text{Total Respondents}} \quad (1)$$

Where:

$\sum$  Respondent Score for PA = Total score of all respondents' answers in one process area.  
 Number of Respondents = Total individuals who answered the questionnaire

### B. GAP Analysis

These GAP percentages are categorised into several assessment levels. Table 6 presents the GAP Analysis categories used to assess the conformity of each process area with CMMI standards. GAP percentages of 20% or below indicate a Good level, meaning the process is functioning effectively. Scores between 21% and 40% fall under Fair, where most activities are implemented but still require refinement. A GAP of 41%–60% is categorised as Needs Improvement, showing many aspects are not aligned with standards. GAP values above 60% are considered Poor, indicating critical deficiencies in process implementation.

**Table 6.** GAP Analysis

| GAP Percentage (%) | Assessment Category | Interpretation                                                                                          |
|--------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| $\leq 20\%$        | Good                | The process has been running well and according to CMMI standards.                                      |
| 21% - 40%          | Fair                | Most of the processes have been implemented, but there are still shortcomings that need to be improved. |
| 41% - 60%          | Needs Improvement   | Many aspects are not in line with CMMI standards and need to be improved.                               |
| $> 60\%$           | Poor                | Processes are not well implemented and require major improvements.                                      |



The calculation of the gap value is performed using the GAP Formula, as shown in Equation 2.

$$\text{GAP} = 100\% - \left( \frac{\text{Average Score}}{\text{Maximum Score}} \times 100\% \right) \quad (2)$$

Where

Average Score: The average result of the assessment for each process area.

Maximum Score: The highest possible score (in this Likert scale is 5).

### C. GAP Results

Table 7 summarises the GAP results for each assessed process area, showing the distance between current implementation and CMMI-DEV 1.3 standards. Project Planning (PP) and Project Monitoring and Control (PMC) both achieve a 20% gap, indicating good performance. Configuration Management (CM) records a 40% gap, reflecting a fair level of implementation. Requirements Management (REQM) and Process and Product Quality Assurance (PPQA) show higher gaps of 53% and 43%, categorised as needing improvement. The largest gaps, 80%, occur in Measurement and Analysis (MA) and Supplier Agreement Management (SAM), indicating poor implementation.

**Table 7.** GAP Results

| Process Area                                 | Average GAP (%) | Description       |
|----------------------------------------------|-----------------|-------------------|
| Project Planning (PP)                        | 20%             | Good              |
| Requirements Management (REQM)               | 53%             | Needs Improvement |
| Process and Product Quality Assurance (PPQA) | 43%             | Needs Improvement |
| Project Monitoring and Control (PMC)         | 20%             | Good              |
| Configuration Management (CM)                | 40%             | Fair              |
| Measurement and Analysis (MA)                | 80%             | Poor              |
| Supplier Agreement Management (SAM)          | 80%             | Poor              |

## Discussion

The findings of this study indicate heterogeneous levels of process maturity across the seven evaluated process areas, reflecting both established practices and substantial deficiencies in the website management and development processes at UNJANI. The relatively low gaps observed in Project Planning (PP) and Project Monitoring and Control (PMC), each at 20%, suggest that fundamental project management mechanisms are already in place. This result aligns with prior studies demonstrating that organizations at early maturity levels often prioritize scheduling, resource allocation, and progress tracking as foundational competencies [5], [9]. Consistency in these areas indicates partial compliance with CMMI-DEV 1.3 expectations and provides a stable basis for further process improvement [12].

In contrast, the substantial gap identified in Requirements Management (REQM) (53%) highlights weaknesses in requirements elicitation, documentation, and change control. Inadequate requirements management has been widely recognized as a critical risk factor leading to scope creep and inconsistent development outcomes [13]. Similar maturity assessments in Indonesian organizations have reported comparable challenges, particularly in aligning stakeholder needs with technical specifications [8], [18]. These findings emphasize the need for more systematic requirement validation and traceability mechanisms.

Process and Product Quality Assurance (PPQA), with a gap of 43%, further indicates that quality-related activities are not yet institutionalized. Partial implementation of audits and reviews may reduce the organization's ability to ensure process compliance and continuous improvement. Previous case studies suggest that insufficient PPQA practices often limit the effectiveness of broader software process improvement initiatives [17]. Strengthening PPQA would therefore contribute to more reliable and repeatable outcomes.

Configuration Management (CM) shows a moderate gap (40%), reflecting a basic but incomplete implementation of version control and change management. This finding is consistent with earlier studies noting that CM practices in academic or semi-formal environments tend to be reactive rather than systematic [9]. Improved CM discipline is essential to support process stability as organizational complexity increases.

The most critical deficiencies are observed in Measurement and Analysis (MA) and Supplier Agreement Management (SAM), each with an 80% gap. The absence of robust measurement frameworks limits evidence-based decision-making and hinders long-term process optimization [15]. Similarly, weak supplier management poses risks to accountability and service continuity, particularly in environments reliant on external vendors [10]. Overall, these results reinforce the need to prioritize MA and SAM as strategic enablers for advancing maturity and achieving sustainable process improvement within the CMMI framework [23].

## Conclusion

Based on the assessment and analysis conducted in this study, it can be concluded that the maturity level of the UNJANI website remains largely in the Fair category, with several key process areas requiring significant improvement. Although Project Planning (PP) demonstrates strong performance with a 20% gap and falls under the Good category, other critical areas such as Requirements Management (REQM) and Process and Product Quality Assurance (PPQA) still show notable deficiencies, indicating the need for further development. The overall evaluation of the software development process using the CMMI-DEV 1.3 framework also reveals that the website has not yet met the criteria for Maturity Level 2, which requires consistent implementation across seven process areas. As a result, the maturity of the UNJANI website is determined to be at Level 1, especially considering that some areas, such as Supplier Agreement Management (SAM) and Measurement and Analysis (MA), are less relevant or insufficiently implemented in the current organisational context. To enhance the overall maturity level, improvements are necessary in areas such as configuration management, documentation and control of user requirements, and thorough feature validation. Strengthening these aspects will help move the website development process toward higher maturity and better alignment with CMMI standards.

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